

Joshua Payapulli

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EDUCATION

University of Southern California

M.S. Applied Data Science

Relevant classes: Data Mining, Machine Learning, Databases, Optimization, UX design and strategy, Fairness in AI

Los Angeles, CA

Aug 2023 - May 2025

GPA: 3.88

The University of Manchester

B.S. Computer Science.

Relevant classes: Fundamentals of AI, Algorithms, Machine Learning and Optimization, etc

Manchester, United Kingdom

Sept 2018 - June 2021

Grade: First Class Honors

EXPERIENCE

Averlon

Software Engineer Intern

United States (Remote)

July 2025 - Oct 2025

- Developed Python pipelines to transform Terraform configurations into inputs for LLM-driven risk analysis, applying RAG and in-context learning to generate blast-radius forecasts and automatically tag high-risk cloud pull requests.
- Designed agentic workflows integrating LLMs to reason over IAM policies across AWS and Azure, generating structured, explainable insights that proactively strengthen multi-cloud security.
- Modeled attack-chain graphs as WCSPs and implemented min-cut optimization to identify minimal-cost mitigation paths; built an adjacency-matrix reachability algorithm to surface exposed assets within constrained IP regions.

Align Technology

Data Analytics Intern

San Jose, CA

June - Aug 2024

- Developed a scalable document-clustering pipeline using K-means and cosine similarity to detect and remove duplicate policy documents, significantly improving retrieval speed and reducing corpus size by 70%.
- Built data processing workflow in Python for ingesting and transforming compliance data into Azure Data Lake, enabling downstream ML-driven document classification and analytics.
- Implemented automated validation and logging to monitor data integrity and ensure reliable inputs for ML workflows.

AND Digital

Full-stack Software Engineer

Leeds, United Kingdom

Oct 2022 - July 2023

- Developed real-time tracking and event-management features using React Native, Redux, and Firebase, improving team visibility and coordination for over 100 employees across offices.
- Designed and implemented RESTful APIs to manage event subscriber notifications, enabling real-time updates, reducing delivery latency by over 30%, and ensuring seamless integration between mobile and web clients.
- Integrated event logging and analytics pipelines to track user interactions and monitor app performance in real time.

PricewaterhouseCoopers (PwC)/ First Abu Dhabi Bank

Consultant/ Business Analyst - Contract

Dubai, United Arab Emirates

Aug 2021 - July 2022

- Built data cleansing and transformation scripts to maintain high data integrity for compliance reporting.
- Automated compliance-reporting and forecasting logic, improving data accuracy and cutting SteerCo report turnaround by 50%.

Careys Plc

Data Analyst Intern

London, United Kingdom

June 2019 - Aug 2019

- Built ETL pipelines in Python and SQL to integrate weather data into Azure for safety-risk tracking at construction sites.

PERSONAL PROJECTS

Stryde – AI-Powered Personal Running Coach

- Developed an AI agent that analyzes Strava data to detect training patterns, dynamically generating adaptive workout plans.
- Built a RAG pipeline with FastAPI and FAISS for personalized updates and deployed the app with automated CI/CD on Vercel.

PlanLA – LA 2028 Olympic Impact Simulator

- Built a real-time geospatial model of LA 2028 investment impact using spatial joins and weighted displacement-risk scoring.
- Investment impact spreads by distance and centrality, with an AI assistant recommending strategies from the live simulation.

TECHNICAL SKILLS

Languages: Python (PyTorch, TensorFlow, scikit-learn), SQL, Go, JavaScript, TypeScript, Java

Tools/Frameworks: AWS, Azure, Databricks, Terraform, Docker, Kubernetes, Tableau, Power BI, Git/GitHub

Other key competencies: LLM Integration, Agentic AI, Retrieval-Augmented Generation (RAG), In-Context Learning (ICL), Fine-Tuning & Evaluation, ETL & Data Pipeline Engineering, Cloud Data Infrastructure, Model Deployment, Scalable Systems Design